EXPERIMENT 4

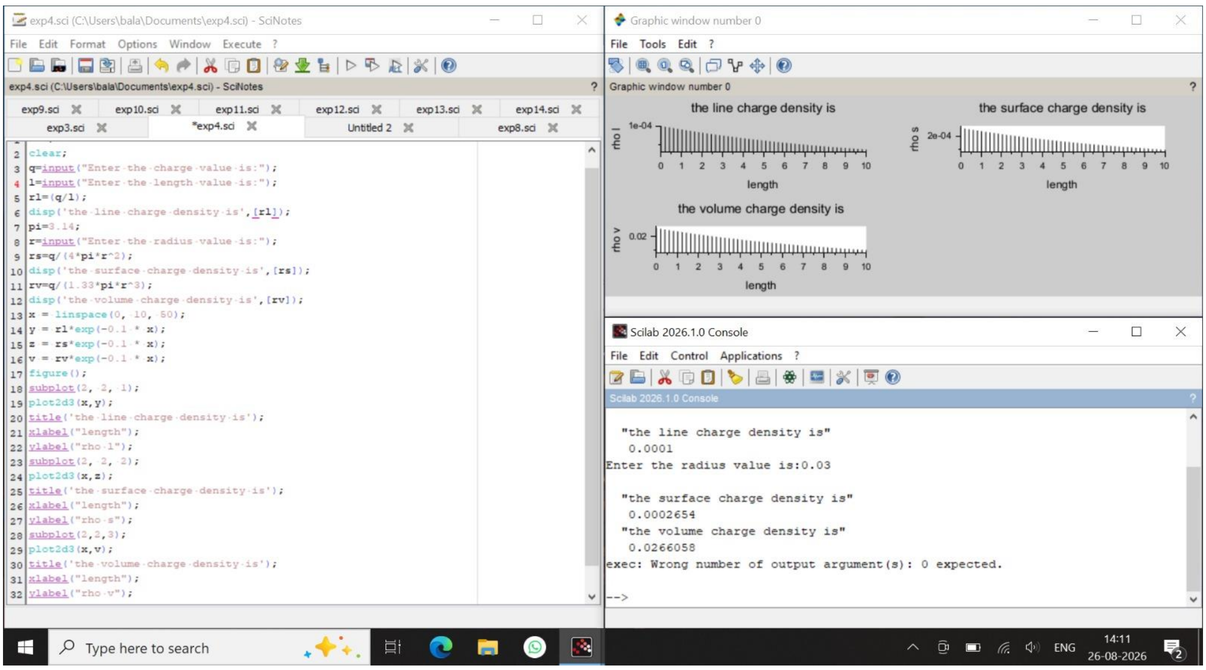
Analysis of Continuous Charge Distribution in electric field. To find the (i) line charge density, (ii) Surface charge density and (iii) Volume charge density for any specific (a) line, (b) surface or (c) volume.

Aim: To write program for continuous Charge Distribution in electric field. The different types of charge density there are (a) Linear Charge Density, (b) Surface Charge Density, (c) Volume Charge Density using SCILAB.

Software required:

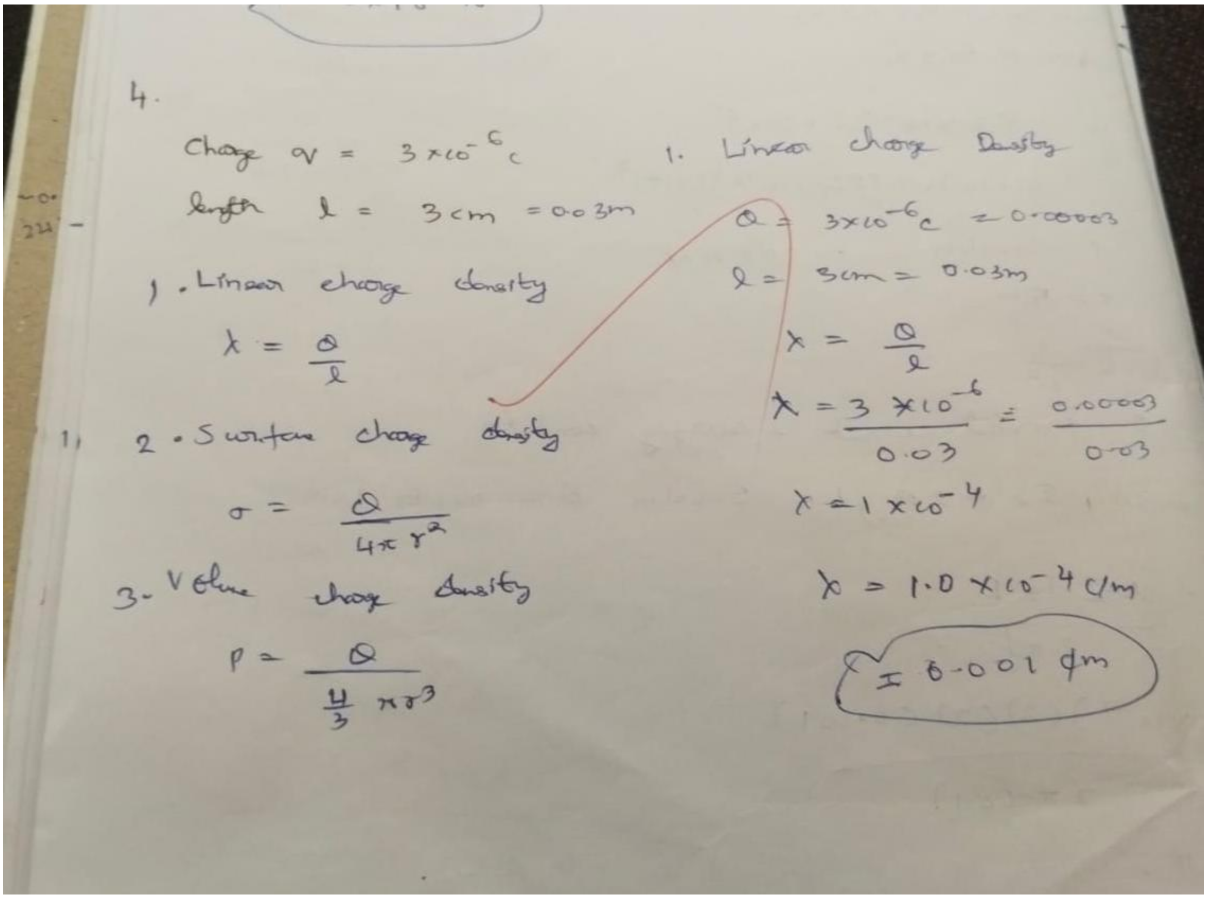
•SCILAB version 6.0.1

Program:



Output:

Theoretical Calculation :



4.

Charge $q = 3\times10^{-6}\,C$

length $l = 3\,cm = 0.03\,m$

1. Linear charge density

$$\lambda = \frac{Q}{l}$$

2. Surface charge density

$$\sigma = \frac{Q}{4\pi r^2}$$

3. Volume charge density

$$\rho = \frac{Q}{\frac{4}{3}\pi r^3}$$

1. Linear charge Density

$Q = 3\times10^{-6}\,C = 0.000003$

$l = 3cm = 0.03m$

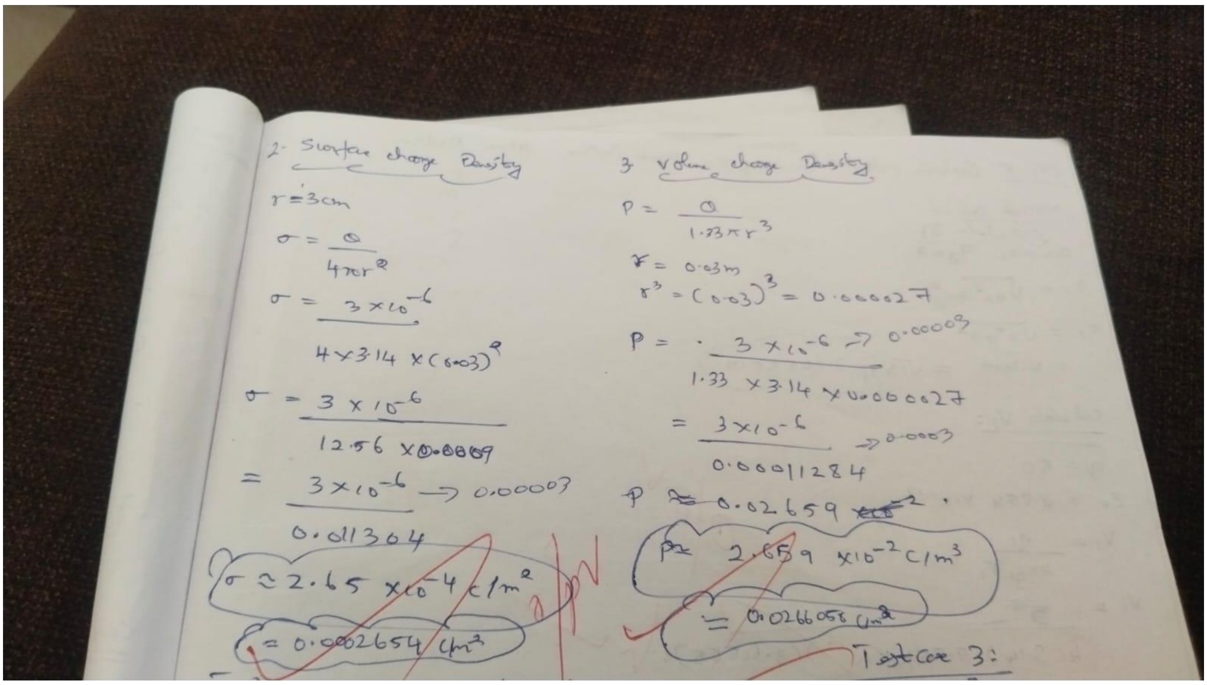
$$\lambda = \frac{Q}{l}$$

$$\lambda = \frac{3\times10^{-6}}{0.03} = \frac{0.000003}{0.03}$$

$\lambda = 1\times10^{-4}$

$\lambda = 1.0\times10^{-4}\,C/m$

$= 0.001\,C/m$



2. Surface charge Density

$r = 3cm$

$$\sigma = \frac{Q}{4\pi r^2}$$

$$\sigma = \frac{3\times10^{-6}}{4\times3.14\times(0.03)^2}$$

$$\sigma = \frac{3\times10^{-6}}{12.56\times0.0009}$$

$$= \frac{3\times10^{-6}}{0.011304} \rightarrow 0.000003$$

$\sigma \simeq 2.65\times10^{-4}\,C/m^2$

$= 0.0002654\,C/m^2$

3. Volume charge Density

$$\rho = \frac{Q}{1.33\pi r^3}$$

$r = 0.03m$

$r^3 = (0.03)^3 = 0.000027$

$$\rho = \frac{3\times10^{-6}}{1.33\times3.14\times0.000027} \rightarrow 0.000003$$

$$= \frac{3\times10^{-6}}{0.00011284} \rightarrow 0.0003$$

$\rho \approx 0.02659\ \cancel{\times10^{-2}}$

$\rho = 2.659\times10^{-2}\,C/m^3$

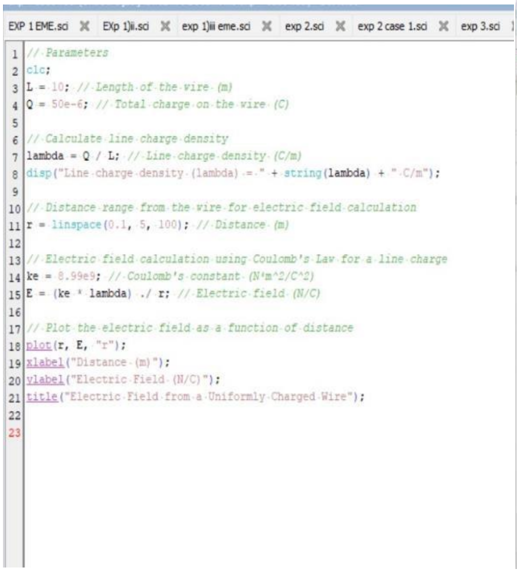
$= 0.0266058\,C/m^3$

Test case 3:

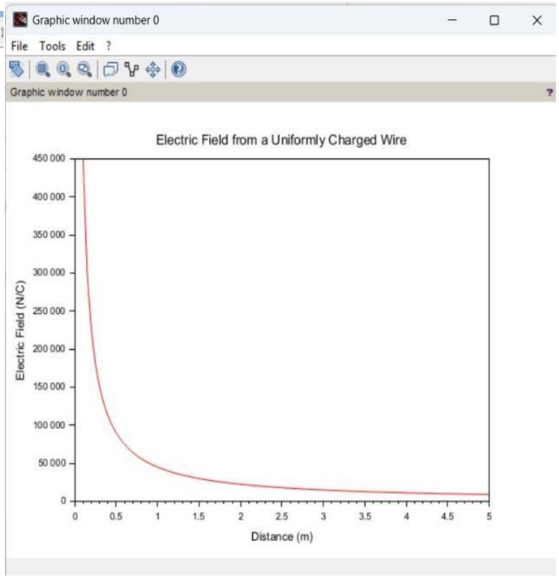
Case 1:

A long wire of length L=10m carries a uniform charge of Q=50µC. The goal is to calculate the line charge density λ and determine how the electric field behaves as a function of distance from the wire.

Program: **Output:**



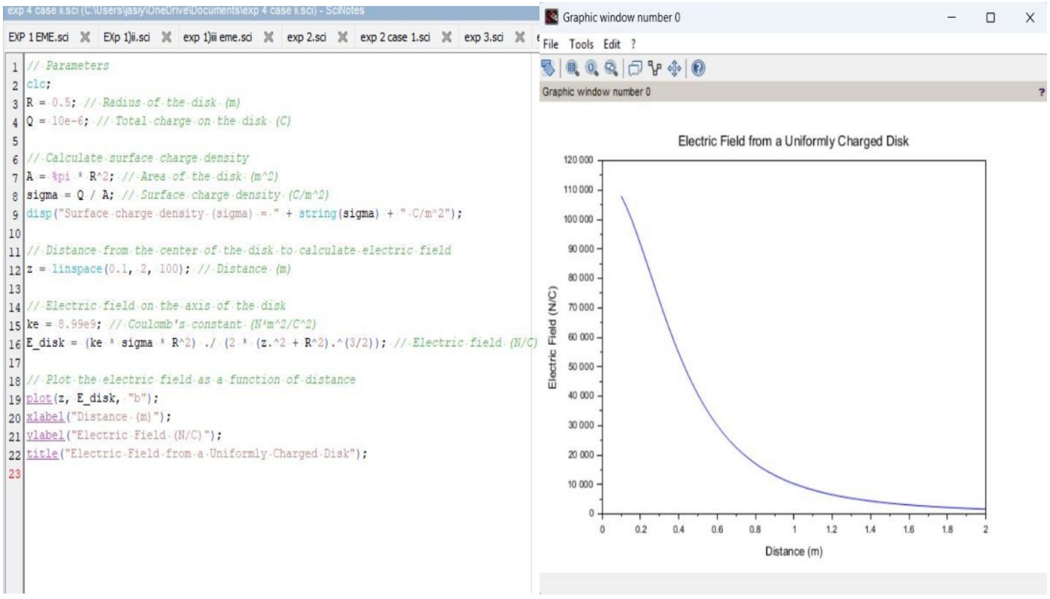
```
// Parameters
clc;
L = 10; // Length of the wire (m)
Q = 50e-6; // Total charge on the wire (C)

// Calculate line charge density
lambda = Q / L; // Line charge density (C/m)
disp("Line charge density (lambda) = " + string(lambda) + " C/m");

// Distance range from the wire for electric field calculation
r = linspace(0.1, 5, 100); // Distance (m)

// Electric field calculation using Coulomb's Law for a line charge
ke = 8.99e9; // Coulomb's constant (N*m^2/C^2)
E = (ke * lambda) ./ r; // Electric field (N/C)

// Plot the electric field as a function of distance
plot(r, E, "r");
xlabel("Distance (m)");
ylabel("Electric Field (N/C)");
title("Electric Field from a Uniformly Charged Wire");
```



Case 2:

Given a uniformly charged disk of radius R=0.5m with total charge Q=10µC, calculate the surface charge density and plot the electric field along the axis of the disk. Explain the behavior of the electric field as a function of distance from the center of the disk.

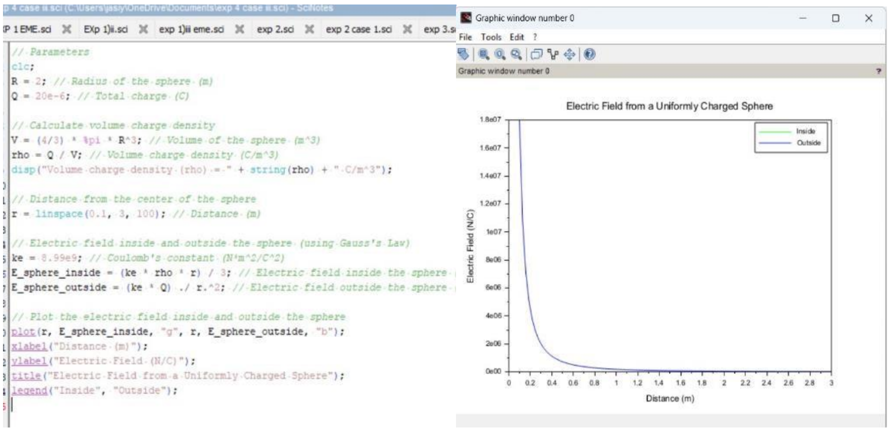
Code: **Output:**



Case 3:

Given a uniformly charged sphere of radius R=2m with total charge Q=20μC, calculate the volume charge density and plot the electric field both inside and outside the sphere. Explain the difference in electric field behavior for points inside and outside the sphere.

Code: **Output:**



Result : Thus the program was verified for continuous Charge Distribution in electric field. The different types of charge density there are (a) Linear Charge Density, (b) Surface Charge Density, (c) Volume Charge Density using SCILAB was successfully.